

21.11.23 §1. Simple and Klein sing's

germs $(\mathbb{C}^n, 0) \rightarrow (\mathbb{C}, 0)$

1.

Def A class of singularities is a subset $K \subseteq \mathcal{O}_n$ s.t. it is invariant for $\text{Diffeo}((\mathbb{C}^n, 0))$.

Def A normal form for a class K is a map $\phi: B \rightarrow M$, where B is a vector space $M = \mathbb{C}[x_1, \dots, x_n]$ s.t.

1. $\phi(B)$ intersects all orbits in K
2. For each orbit $A \in K$ $\phi^{-1}(A)$ is finite
3. $\phi^{-1}(M \setminus K)$ is contained in a hypersurface.

It is simple if ϕ has the form $\phi(b_1, \dots, b_r) = \phi(x) + \sum_{i=1}^r b_i x_i^{m_i}$ for a fixed polynomial $\phi(x)$.

$x^3 + y^3 + z^3 + xyz$ parabolic

Simple sing's:

$A_{k \geq 1}$ x^{k+1}	D_k $k \geq 4$ $x^2 y + y^{k+1}$	E_6 $x^3 + y^4$	E_7 $x^3 + xy^3$	E_8 $x^3 + y^5$
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There is a correspondence w/ Klein singularities (which are vertices you classify them)

Given by $\Gamma < SO(3)$ finite in the following way:

~~Let $\Gamma < SO(3)$ be a finite subgroup~~

Let $\phi: SU(2) \rightarrow SO(3)$ be the 2:1 cover map

Γ

Let $\Gamma' = \phi^{-1}(\Gamma)$

Then

$\Gamma' \subset \mathbb{C}^2$

and

$\mathbb{C}[x, y]^{\Gamma'} \simeq \frac{\mathbb{C}[a, b, c]}{f(a, b, c)}$

is an hypersurface in $\mathbb{C}^3$ orbit space

Classification of $\Gamma < SO(3)$

1. $\mathbb{Z}_n = \langle \text{rot}(2\pi/n) \rangle \rightsquigarrow A_n \quad x^n + yz = 0$

2. $D_{2n} = \langle \text{symmetries of the regular } n\text{-gon} \rangle \quad xy^2 - x^{n+1} + z^2 = 0$

3. $T_{12} = \text{symmetries of the tetrahedron} \quad E_6 \quad x^4 + y^3 + z^2 = 0$

$O_{24} = \text{symmetries of the octahedron} \quad E_7 \quad x^3 + xy^3 + z^2 = 0$

$I_{60} = \text{symmetries of the icosahedron} \quad E_8 \quad x^5 + y^3 + z^2 = 0$

CORRESPONDENCE: For A_n, D_n, E_n

The zero level set of the singularity

$\simeq V \subset \mathbb{C}^3$ orbit space

§2. ADE singularities and Coxeter/Weyl groups

12

$\mathbb{R}^n$ euclidean space

Def: ~~the reflection~~
Given $v \in \mathbb{R}^n$, the reflection in $H_v := \{x : \langle x, v \rangle = 0\}$ is the
transform $S_v : x \mapsto x - 2 \frac{\langle x, v \rangle}{\langle v, v \rangle} v$. H_v is called the mirror.

Let $W \subseteq \mathbb{R}^n$ be generated by reflections,

Then a connected component of $\mathbb{R}^n \setminus (\cup_{\text{SEW reflection}} H_s)$ is called a chamber.

For a fixed chamber C , ~~a minimal set of mirrors~~ $\{H_i\}$

let $\{H_i\}$ be a minimal set of mirrors s.t. $C = \cap_i H_i^+$. H_i are called walls of C .

Theorem

1. C is a simplicial cone
2. $W = \langle s_{H_i} : H_i \text{ is a wall of } C \rangle$
3. C is a fundamental domain for W
4. $\angle(H_i, H_j)$ is π/k for $k \geq 2$
5. $W = \langle s_i : s_i^2 = 1, (s_i s_j)^{m_{ij}} = 1 \text{ where } \angle(H_i, H_j) = \pi/m_{ij} \rangle$

Coxeter graph: Vertices = $\{H_i\}$ walls of a chamber

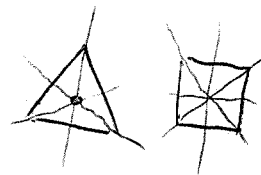
Edges: $\langle i, j \rangle$ has multiplicity k if $m_{ij} = \pi/k$

Convention: don't draw $\langle i, j \rangle$ if $m_{ij} = 2$
no multiplicity if $m_{ij} = 3$

Ex.

$P_n \subseteq \mathbb{R}^2$ regular n -gon

$\leadsto W = \langle s_{H_i} : \text{ref } H_i \text{ hyperplane through midpoints of edges and vertices} \rangle$



Ex. A_n

$S_{n+1} \subseteq \mathbb{R}^{n+1}$ action by permutations $\mathbb{R}^n = \{x_1 + x_2 + \dots + x_{n+1} = 0\} \subseteq \mathbb{R}^{n+1}$ is fixed

$S_{n+1} \subseteq \mathbb{R}^n$ is generated by S_{ij} reflection in $H_i = \{x : x_i = x_j\}$

A chamber is given by $C := \{x : x_1 \leq x_2 \leq \dots \leq x_n\}$ w/ walls $H_i = \{x : x_i = x_{i+1}\}$

Anales $\angle(H_i, H_{i+1}) = \pi/2$ $\angle(H_i, H_j) = \pi/2$ if $|i-j| > 1$

Ex $W \cong W' \iff$ isomorphic graphs

3.

Theorem W finite irreducible reflection group
 $\iff$ its Coxeter graph is of the form:

A_μ  ($\mu \geq 1$ vertices)

B_μ  ($\mu \geq 2$ ")

D_μ

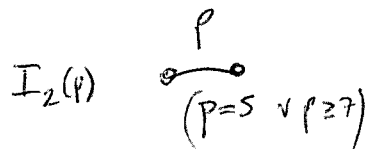
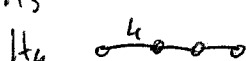
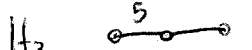
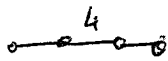
E_6

E_7

E_8

F_4

G_2



Coxeter transformation $h := s_1 \circ \dots \circ s_\mu$ where s_i reflections in the walls of a chamber

Theorem Coxeter transf.'s are all conjugate

- Coxeter transf.'s are all conjugate
- Eigenvalues of h are $\exp(2\pi i/m_j)$ for $0 < m_1 \leq \dots \leq m_\mu < |h|$
 m_j known

e.g. A_μ $1, 2, \dots, \mu$ $\text{ord}(h) = \mu + 1$

- $m_1 = 1$, $m_\mu = \text{ord}(\mu) - 1$, $m_k + m_{\mu+1-k} = \text{ord}(h)$
 $m_j :=$ exponents of W

Example

A_μ

$h = s_1 \circ \dots \circ s_\mu$

corresponds to

$$\begin{pmatrix} 1 & 2 & 3 & \dots & \mu+1 \\ \mu+1 & 1 & 2 & & \mu \end{pmatrix}$$

$$s_i \leftrightarrow (i, i+1): H_i := \{x: x_i - x_{i+1} = 0\}$$

$$s_i: e_i \mapsto e_i - (e_i - e_{i+1}) = e_{i+1}$$

$$e_{i+1} \mapsto e_{i+1} - (e_{i+1} - e_i) = e_i$$

Theorem W Coxeter grp.

$\mathbb{C}[[x]]^W \cong \mathbb{C}[[a_1, \dots, a_\mu]]$ is a free algebra, with a_i homogeneous of degree m_i .

It follows

$$\mathbb{C}^\mu \xrightarrow{\pi} \mathbb{C}^\mu / W \cong \mathbb{C}^\mu$$

is a $|W|$ -cover

$$\text{branched along } \pi \left(\bigcup_{H_i \text{ minor}} H_i \right) =: \sum_{\text{swallowtail}} (W)$$

π is called the Vieta map

Example A_μ $S_\mu \subset \mathbb{C}^{\mu+1}$

invariants are the elementary symmetric functions

$$\lambda_i = (-1)^k \sum_{j_1 \neq j_2 \neq \dots \neq j_k} x_{j_1} x_{j_2} \dots x_{j_k} \quad k = \mu - i + 1$$

$$\deg(\lambda_i) = \mu + 1 - i \quad \text{for } i = 0, \dots, \mu$$

The orbit of $\underline{x} = (x_1, \dots, x_{\mu+1}) \in \mathbb{C}^{\mu+1}$ is identified w the unordered set $\{x_1, \dots, x_{\mu+1}\}$

~~and~~ ~~is characterized~~ i.e. the roots of

$$\prod_{i=1}^{\mu+1} (t - x_i) = t^{\mu+1} + \lambda_{\mu-1} t^\mu + \dots + \lambda_1 t + \lambda_0$$

$$\text{For } A_2 \quad \underline{x} \in \mathbb{C}^3 = t^3 + \lambda_1 t + \lambda_0$$

$$\text{so that } \Sigma(A_2) = \{4\lambda_1^3 + 27\lambda_0^2 = 0\}$$

Root systems & Weyl Groups

Def A finite set $0 \neq R \subseteq \mathbb{R}^m$ s.th. $\langle R \rangle_{\mathbb{R}} = \mathbb{R}^m$ is a root system if

$$i) \quad \forall \alpha \in R \quad s_\alpha(R) = R \quad (\text{stable under reflections in } R)$$

$$ii) \quad \forall \alpha, \beta \in R \quad s_\alpha(\mathbb{Z}\alpha + \mathbb{Z}\beta) = \mathbb{Z}\alpha + \mathbb{Z}\beta$$

$$\text{Obs: } (ii) \iff \frac{2\langle \alpha, \beta \rangle}{\langle \alpha, \alpha \rangle} \in \mathbb{Z} \quad \text{indeed } s_\alpha(\alpha) = -\alpha$$

$$s_\alpha(\beta) = \underbrace{\beta - 2 \frac{\langle \alpha, \beta \rangle}{\langle \alpha, \alpha \rangle} \alpha}_{\in \mathbb{Z}\alpha + \mathbb{Z}\beta}$$

$$\text{reduced: } \iff \mathbb{R} \cdot \alpha \cap R = \{\pm \alpha\}$$

Given a chamber for $W(R)$ Weyl group

$$C = \cap_{i=1}^r H_i^+$$

H_i walls

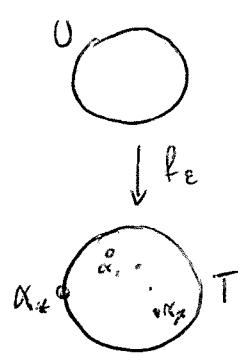
the elements α_i s.th. $\langle \alpha_i, C \rangle \geq 0$ are called a basis.

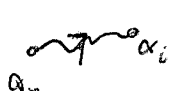
Theorem If $\alpha_1, \dots, \alpha_r$ is a basis of R , then $\beta = \sum a_i \alpha_i$ w either $a_i \geq 0 \quad \forall i$ or $a_i \leq 0 \quad \forall i$

Dynkin diagram: Vertices = $\{\alpha_i\}$ basis
edges of multiplicity k if $\angle(\alpha_i, \alpha_j) = \pi \left(\frac{k+1}{k+2} \right)$

oriented $\alpha_i \rightarrow \alpha_j$ if $\|\alpha_i\| > \|\alpha_j\|$

Recap: Given $f: (\mathbb{C}^n, 0) \rightarrow (\mathbb{C}, 0)$ isolated sing,
 there is a ~~well~~ morsification w/ μ singular points



$H_{n-1}(V_x)$ is generated by the vanishing cycles Δ_i associated to WD paths 

$h_x: H_{n-1}(V_x) \rightarrow H_{n-1}(V_x)$
 $\delta \mapsto \delta + (\delta, \Delta) \cdot \Delta$
 is the classical monodromy operator
 is induced by $\gamma = \partial T$

Thm

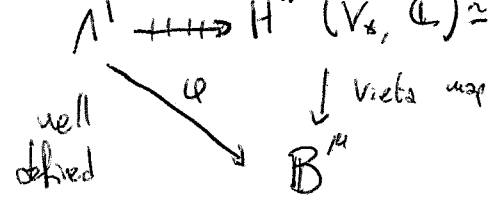
If f is a simple sing., then $(\Delta_i)^2 = -2$
 so $H_{n-1}(V_x, \mathbb{Z}) \subseteq H_{n-1}(V_x, \mathbb{R}) \simeq \mathbb{R}^\mu$ is a root lattice
 and the monodromy is identified w/ $w(\mathbb{Z})$.

~~To a singularity of type A~~
 f of type A, D, E $\Rightarrow w(\mathbb{Z})$ of type ADE.
 classical monodromy $h_{\partial T} \mapsto$ Coxeter transform.

~~Let $T' = T \setminus \{\alpha_1, \dots, \alpha_\mu\}$~~

Let $\Lambda \subseteq \mathbb{C}^\mu$ the basis of the uni-versal defo
 and $\Lambda' = \Lambda \setminus \Sigma$ w/ Milnor fibration $V' \rightarrow \Lambda'$

~~H_{n-1} be the homology~~
 H^{n-1} the cohomology bundle.
 A section $\sigma \in H^{n-1}$ induces a multivalued map
 $\Lambda' \dashrightarrow H^{n-1}(V_x, \mathbb{C}) \simeq \mathbb{C}^\mu$ w/ well-def. to monodromy



Theorem

φ extends to an iso
 $\Lambda \rightarrow B^\mu$ which restricts to an
 iso $\Sigma \rightarrow \Sigma(w)$